

The Emerging Era of Molecular Medicine

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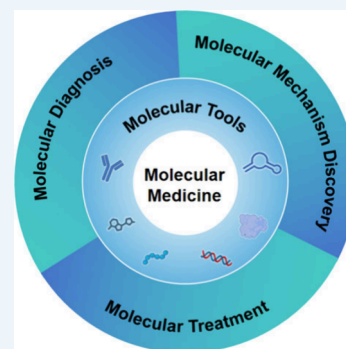
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ABSTRACT: The era of molecular medicine arose as we began to diagnose and treat diseases based on understanding how genes, proteins, and cells work, providing optimal therapeutic care through molecular profiling. Central to molecular medicine is molecular recognition, which is underpinned by techniques involving omics analysis, gene editing, and targeted agents. Recent advancements in these tools not only expand our understanding of biological processes but also aid in the development of diagnostic and treatment modalities at the molecular level, thus bridging the gap between medical research and clinical applications. This perspective traces the development of molecular tools, highlighting, along the way, their pivotal role in advancing molecular medicine for the global health of people.

KEYWORDS: molecular medicine, molecular tools, precision medicine, disease diagnosis, disease treatment, targeted drug, targeted delivery, molecular subtyping



THE HUMAN BODY FUNCTIONS AS A MOLECULAR MACHINE

The evolution of medicine spans millennia, transitioning from macroscopic to molecular levels.¹ Ancient anatomy began before the time of Galen, but it was the advent of microscopic anatomy, centuries later, that expanded our understanding of organs and cells,^{2,3} laying the foundation for modern medicine. Since the 1950s, accelerated research and the decoding of biomolecules such as deoxyribonucleic acid (DNA), proteins, and lipids have deepened our understanding of human biology and physiology as we entered the molecular era.⁴

Cells, the smallest units retaining life properties, are composed of molecules, such as nucleic acids, proteins, carbohydrates, and lipids, forming the basic structural and functional units of organisms. The human body contains approximately 50 trillion cells, with each cell containing 23 pairs of chromosomes and 3.2 billion base pairs,⁵ encoding 20,000–25,000 proteins (Figure 1). The human body functions as a finely tuned “molecular machine”, with every physiological and biochemical process having a molecular basis. For example, postexercise muscle soreness stems from the accumulation of lactate molecules,⁶ perception of temperature is mediated by transient receptor potential (TRP) ion channels,⁷ piezo type mechanosensitive ion channel component 2 (PIEZO2) channels enable touch perception and body position sensing,⁸ and the secretion of dopamine and serotonin contributes to feelings of happiness.^{9,10}

FROM MOLECULAR DISEASES TO MOLECULAR MEDICINE

Aberrations in individual molecules or the molecular network can precipitate the onset and progression of diseases. For instance, sickle cell anemia (SCA) arises from a point mutation in the hemoglobin beta gene (HBB) found on chromosome 11p15.5, leading to the formation of sickle hemoglobin that replaces normal hemoglobin.¹¹ Pauling pioneered the concept of “molecular medicine” by elucidating this mechanism, making a significant shift in disease understanding through a molecular perspective. Discovery of the DNA double helix in the mid-20th century and the establishment of the central dogma of genetic information transfer gave rise to molecular biology, inaugurating an epoch in the life sciences dedicated to investigating life and its processes at the molecular level.^{12,13} In 1994, Kapp and Broad’s review, “New Directions in Molecular Medicine”, formally introduced the concept of “molecular medicine”.¹⁴

Molecular medicine entails dissecting disease mechanisms at the molecular level, enabling early and precise disease diagnosis, and effectively treatment and prevention.¹⁵ The “Human

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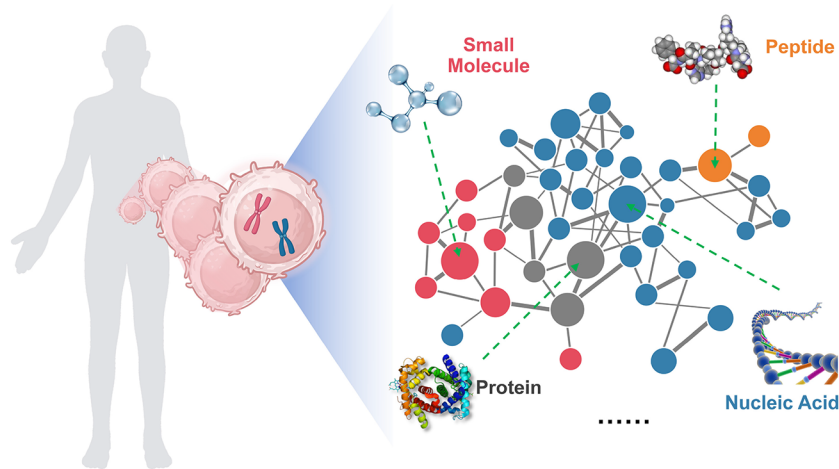


Figure 1. Molecular composition and molecular interaction network of the human body.

Table 1. Comparison of Several Major Molecular Recognition Tools

	Antibody	Peptide	Aptamer	refs
Molecular weight	150 kDa	<5 kDa	5–30 kDa	
Affinity	High (pM–nM)	Low (nM–μM)	High (pM–nM)	
Targets	Immunogenic molecules	Rich targets	Wide range of targets	
Controlled chemical modification	Restricted and uncontrollable	Convenient and controllable	Convenient and controllable	
Internalization into cells	Difficult	Easy	Easy	
Tissue penetration	Poor permeability	Good permeability	Good permeability	
<i>In vivo</i> half-life	Long (~3 weeks)	Short (~hours)	Short (~20 min)	
Serum stability	·Susceptible to temperature	·Susceptible to temperature	·Resistant to high temperature	
	·Resistant to nucleases	·Resistant to nucleases	·Resistant to proteases	
	·Degradable by proteases	·Degradable by proteases	·Degradable by nucleases	
Immunogenicity	High	Low	Low	
Production and cost	Cell culture or animal, and high-cost	Chemical synthesis, and low-cost	Chemical synthesis, and low-cost	
Generation discovery time	Several months	2–8 weeks	Few hours to months	37–40,42
	·Slow selection process and high-cost	·Unstable under high temperature and proteases	·Easily degraded by nucleases	
	·Difficult in chemically modification	·Relatively low affinity	·Short half-life	
	·Limited targets	·Short half-life		
	·Low cellular endocytosis and tissue penetration			
Limitations	·Unstable under high temperature and proteases			
	·High affinity	·Good permeability and internalization into cells	·Controllable chemical modification	
	·Long half-life	·Controllable chemical modification	·High affinity	
Advantages	·Widely used in clinical practice	·Fast selection process and low-cost	·Wide range of targets	
			·Fast selection process and low-cost	

Genome Project” advanced molecular medicine, revealing that aging,^{16,17} neurological diseases,¹⁸ and cancer^{19,20} could be understood molecularly. Molecular tools and practices have thus transformed disease diagnosis and treatment into molecular domain. As an example, molecular diagnostics, such as DNA sequencing, have emerged as the technology of choice to rapidly and efficiently diagnose neurological disorders,²¹ offering significant advantages over magnetic resonance imaging (MRI) and computed tomography (CT) in the early diagnosis of neurological disorders.^{22,23} These molecular techniques not only shorten the time to diagnosis but also help predict the incidence rate. In addition, molecular drugs have become a cornerstone in the treatment of neurological diseases. Notable

examples include the clinical use of cholinesterase inhibitors for Alzheimer’s disease and levodopa for Parkinson’s disease.^{24,25} Immunotherapy, such as OPDIVO (nivolumab), exemplifies this by inhibiting tumor growth through targeting programmed cell death protein 1 (PD-1) and programmed cell death 1 ligand 1 (PD-L1) interaction, with efficacy depending on high PD-L1 expression,²⁶ rather than the cancer’s origin. Similarly, DS8201, a HER2-targeted antibody-drug conjugate, shows therapeutic efficacy across multiple cancer types, based on human epidermal growth factor receptor 2 (HER2) expression.²⁷ A recent report published in *Nature* advocates for studying diseases from a molecular perspective and developing diagnostic and therapeutic modalities that transcend specific cancer tissue types.²⁸ This

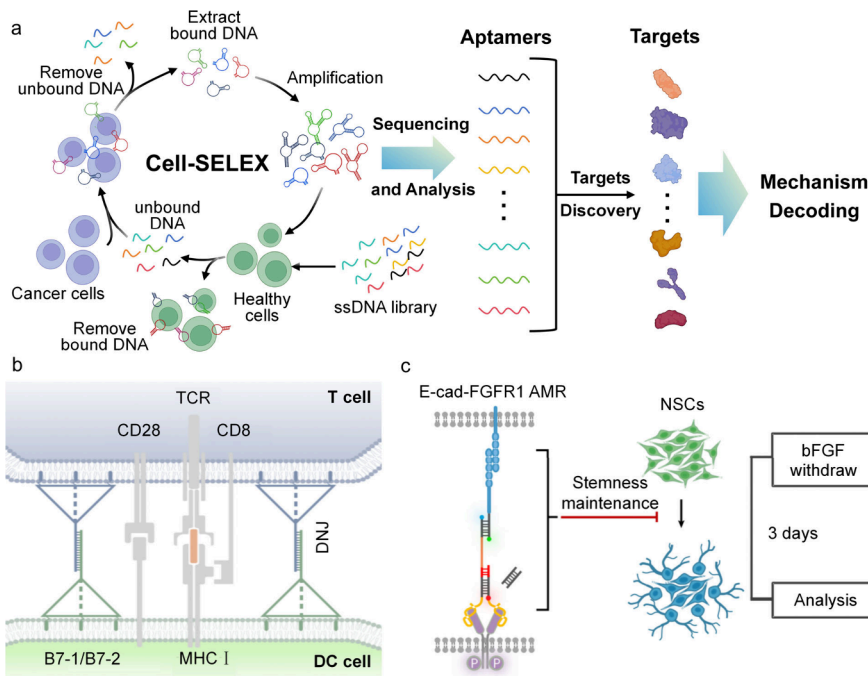


Figure 2. (a) Aptamer Cell-SELEX for discovering cell surface cancer biomarkers. (b) DNA nanotechnology for modulating the APC-T distance to regulate T cell immunity. Reprinted with permission from ref 55. Copyright 2023 Springer Nature. (c) DNA–protein chimera for analyzing intercellular forces in cellular differentiation. Reprinted from ref 56. Copyright 2024 Springer Nature.

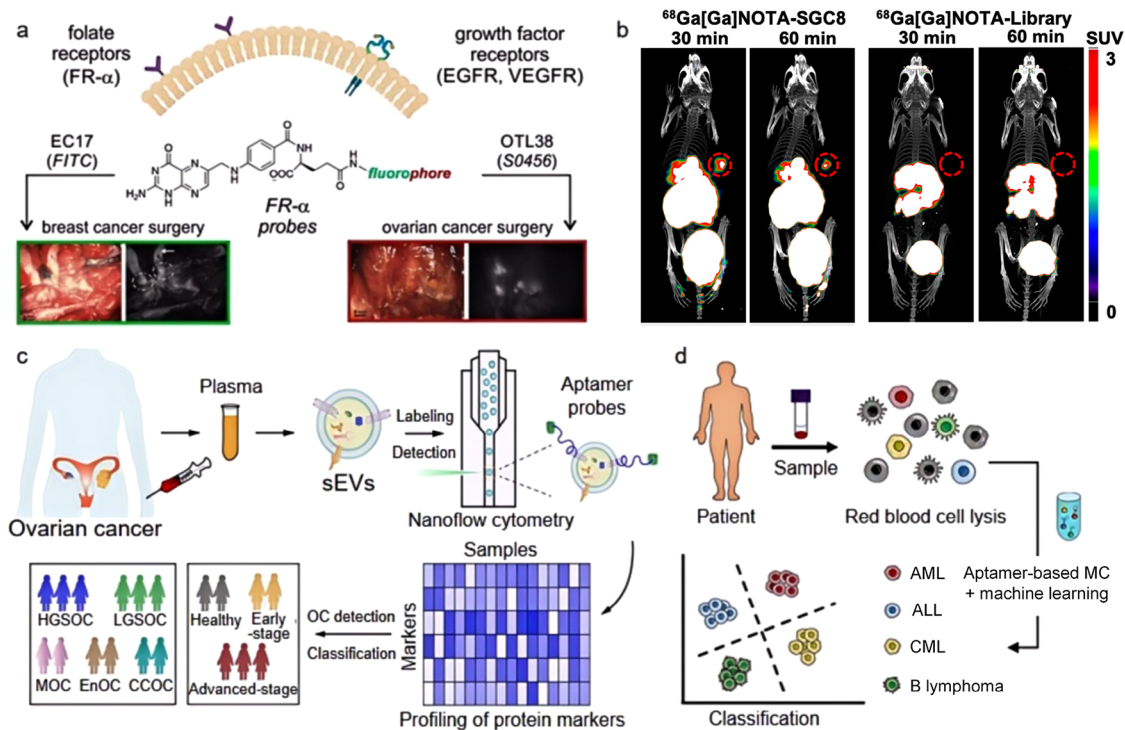


Figure 3. (a) Tumor-targeted fluorescent probe conjugates used in tumor surgery. Reprinted with permission under a Creative Commons CC BY 4.0 license from ref 58. Copyright 2023 American Chemical Society. (b) Positron emission tomography (PET) of the aptamer-radiolabeled conjugate ⁶⁸Ga-NOTA-Sgc8. Reprinted with permission under a Creative Commons CC BY 4.0 license from ref 59. Copyright 2023 American Association for the Advancement of Science. (c) Multiple aptamer-mediated precise diagnosis and cancer subtyping using tumor exosomes. Reprinted with permission from ref 60. Copyright 2023 Wiley-VCH GmbH. (d) Multiple aptamers enabled the molecular subtyping of hematological malignancies. Reprinted with permission under a Creative Commons CC BY 4.0 license from ref 61. Copyright 2024 Chinese Chemical Society.

underscores the increasing significance of molecular-based classification and treatment in oncology.

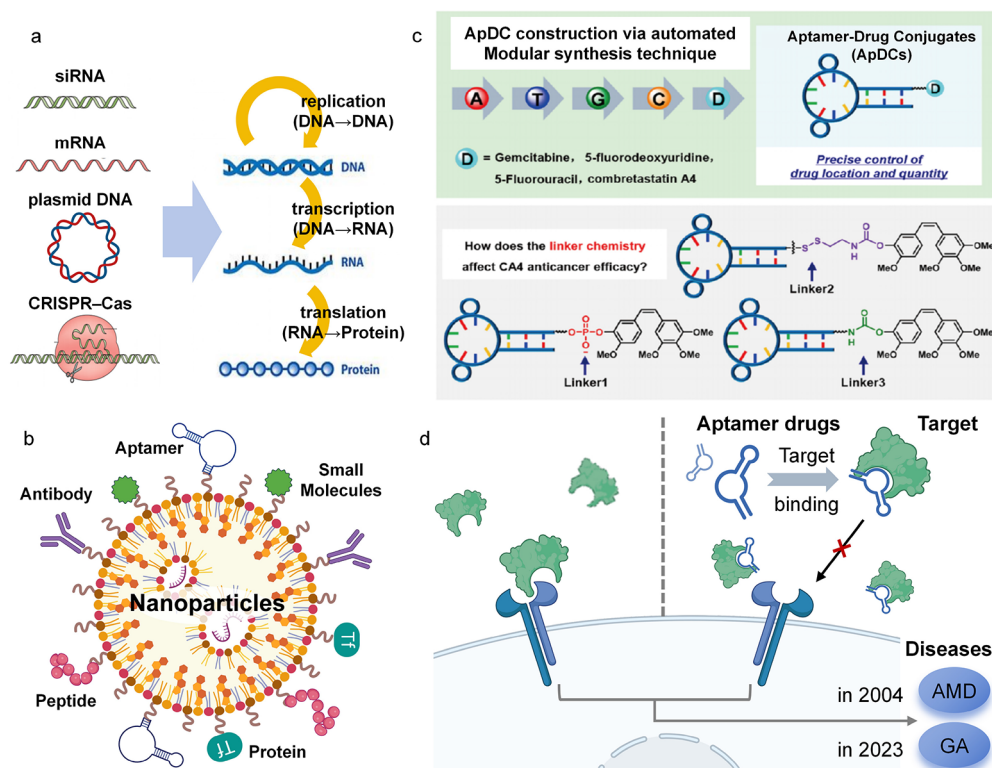


Figure 4. (a) Schematic diagram of gene-regulating molecular tools for intracellular protein expression based on a central dogma. (b) Schematic diagram of molecular tools for LNPs-targeted delivery. (c) Aptamer-drug conjugates for tumor molecular-targeted therapy. Reprinted with permission from ref 71. Copyright 2020 American Chemical Society. (d) FDA-approved aptamer-based drugs for treatment of age-related macular degeneration (AMD) and geographic atrophy (GA).

MOLECULAR TOOLS DRIVING MOLECULAR MEDICINE ADVANCEMENTS

Molecular medicine relies on molecular recognition, and facilitated by innovative tools. These tools, including antibodies,²⁹ peptides,³⁰ aptamers,³¹ and small molecules,³² etc., offer high specificity and affinity in various targets, such as ions,³³ proteins,³⁴ nucleic acids,³⁵ and cells.³⁶ Different molecular tools each have their own advantages and limitations (Table 1). Although antibodies have limitations in targets' richness and uncontrollable chemical modification, they have high affinity and long half-life, making them the most widely used molecular tool in clinical practice.^{37,38} Peptides have limited affinity and *in vivo* half-life, but they have controllable chemical modification and more diverse targets, compared to antibodies.^{37,39} Aptamers, often dubbed "chemical antibodies", are synthesized through systematic evolution of ligands by exponential enrichment (SELEX).⁴¹ Although aptamers are easily degraded by nucleases, they have unique advantages in high affinity, discovering biomarkers, and facilitating molecular programming owing to their screening mode and complementary pairing of nucleic acid bases.^{42,43} Aptamers hold promising potential for revolutionizing disease detection (aptamer sensors), clinical diagnostics, and precision therapeutics. In a word, all advanced molecular tools collectively contribute to significant progress in molecular medicine, particularly in deciphering molecular mechanisms, disease diagnosis, and disease treatment.

Molecular Tools for Deciphering Molecular Mechanism. Understanding disease mechanisms is crucial for precision molecular medicine. Identifying biomarkers and elucidating key proteins' functions and mechanisms in disease pathogenesis are essential. Cell-SELEX presents a particularly

innovative approach for discovering cell surface disease biomarkers⁴⁴ (Figure 2a). To date, Cell-SELEX efforts have yielded thousands of aptamers across various diseases,⁴⁵ spanning more than ten different cancer types, resulting in the identification of several potential targets for tumor therapy, including protein tyrosine kinase 7 (PTK7),^{36,46,49–52} leucine rich pentatricopeptide repeat containing (LRPPRC),^{47,53} and cytoskeleton-associated protein 4 (CKAP4).^{48,54} Furthermore, DNA nanotechnology offers insights into T-cell immune regulation mechanisms by modulating intercellular distance (Figure 2b),⁵⁵ while DNA–protein chimeras offer an opportunity to analyze the impact of intercellular forces on cellular behavior (Figure 2c).⁵⁶

Molecular Tools in Disease Diagnosis. As research continues to reveal the molecular mechanisms underlying diseases, molecular-level diagnostics have simultaneously become integral in guiding clinical treatment strategies. With aptamer-targeting of the nucleocapsid protein (N protein) of coronavirus, rapid at-home self-testing for coronavirus disease 2019 (COVID-19) is now feasible.⁵⁷ Moreover, coupling molecular recognition tools with fluorescent probes or contrast agents illuminates cancer cells, facilitating precise tumor localization and mitigating intraoperative tumor residue (Figure 3a).⁵⁸ Notably, by enhancing diagnostic and therapeutic precision and specificity, the aptamer-radiolabeled conjugate ⁶⁸Ga-NOTA-Sgc8 has demonstrated successful precise imaging of bladder cancer in clinical trials (Figure 3b).⁵⁹ Furthermore, the extensive development of molecular recognition tools holds promise for multiplexed diagnosis, enabling precise cancer subtyping through the analysis of tumor exosomes (EVs)⁶⁰ (Figure 3c) or tumor cells⁶¹ (Figure 3d) with multiple aptamers.

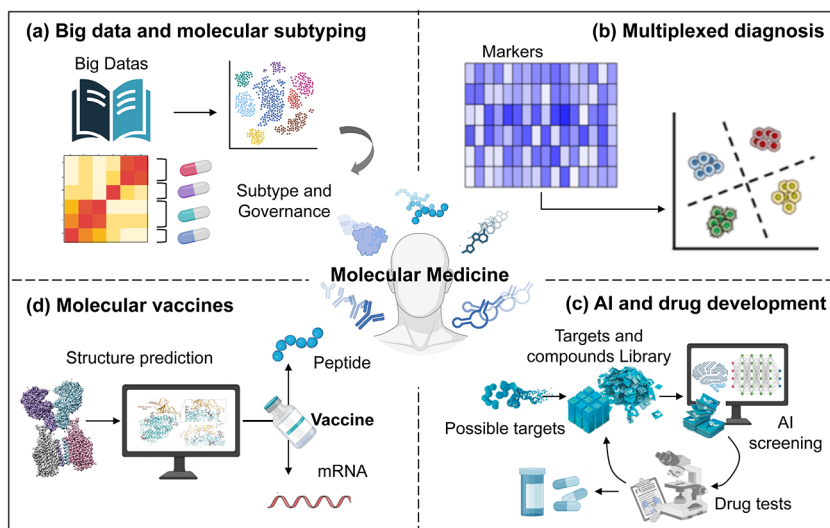


Figure 5. Outlooks and challenges for molecular medicine in the future: (a) Big data and molecular subtyping. (b) Multiplexed diagnosis. (c) AI and drug development. (d) Molecular vaccines.

These advancements lay the foundation for multiplexed disease diagnosis, with future integration of artificial intelligence (AI) expected to boost sensitivity and accuracy.

Molecular Tools in Disease Treatment. Developing molecular tools is essential for achieving precision medicine. Gene-regulating molecular tools like mRNA, siRNA, and CRISPR-Cas have emerged as powerful therapeutic modalities^{62,63} (Figure 4a). mRNA-based vaccines, in particular, have garnered attention for their high potency, rapid development, and cost-effective manufacture,⁶⁴ with lipid nanoparticles (LNPs) encapsulation playing a pivotal role in their success.⁶² Current efforts focus on optimizing LNPs to augment tissue-targeting capabilities via targeted molecular tools for improved disease prevention and therapeutic outcomes^{65–67} (Figure 4b). Additionally, drugs utilizing small molecules, peptides, monoclonal antibodies, and aptamers as targeted molecular tools have been extensively developed.⁶⁸ Among them, antibody-drug conjugates (ADCs) have demonstrated notable success⁶⁹ with 13 Food and Drug Administration (FDA) approvals for the treatment of diseases such as leukemia and breast cancer.⁷⁰ Aptamer-drug conjugates also show promise as effective molecular-targeted therapies (Figure 4c), with several candidates in preclinical and clinical trials exhibiting encouraging efficacy.^{71–74} Furthermore, the pursuit of functional nucleic acid drugs represents another significant frontier in molecular-targeted therapy (Figure 4d) with two aptamer-based drugs already FDA-approved and others advancing through clinical trials.^{75–78}

STUDY LIMITATIONS

Although molecular medicine has made significant achievements, its diagnosis and treatment of complex diseases still face enormous challenges. Many diseases are driven by intricate mechanisms involving multiple genes, proteins, and their complex interactions. While molecular medicine can explore these mechanisms, fully understanding the multifactorial interactions and complex networks is still difficult, limiting diagnostic and therapeutic effectiveness.⁷⁹ Although personalized medicine is a key focus, the complexity of the genome and environmental influences often result in varied responses to the same treatment, making it challenging to accurately predict

disease outcomes and treatment efficacy. Additionally, translating basic research into clinical applications is a lengthy and complex process, with many molecular discoveries struggling to be rapidly and effectively applied in clinical settings, particularly for diseases with multiple molecular targets such as cancer. Nevertheless, with the continuous advancement of high-throughput sequencing technologies and artificial intelligence, molecular medicine is gradually overcoming these obstacles. These innovations are enhancing our ability to process and analyze vast amounts of biological data, leading to more precise diagnostics, targeted therapies, and improved prediction of treatment responses, ultimately bringing us closer to personalized and effective interventions for complex diseases.

OUTLOOK

Molecular medicine entails understanding life from a molecular perspective and utilizing molecular tools for the diagnosis, treatment, and prevention of diseases. The future trajectory of molecular medicine includes but is not limited to the following directions (Figure 5):

1. **Big data and molecular subtyping:** Developing molecular tools for high-throughput sequencing and characterization of proteins and metabolites at the single-cell level enables analysis of cellular heterogeneity, facilitating personalized treatment.
2. **Multiplexed diagnosis:** Similar to facial recognition technology, multiplexed diagnosis would leverage the diverse molecular characteristics of a cell to enhance the specificity and sensitivity of disease diagnosis.
3. **AI and drug development:** Researchers are currently engaged in developing machine learning algorithms to analyze disease biomarkers, predicting their structures, and functions. AI aid in molecular tool design and screening would optimize the drug development process, propelling the biopharmaceutical field toward swifter, more precise, and safer outcomes.
4. **Molecular vaccines:** The future developmental trajectory of molecular vaccines encompasses tailored designs for various diseases, the application of gene editing technology to boost immune effects, and the utilization

of nanotechnology to enhance vaccine delivery efficiency and stability.

These directions represent the ongoing evolution of molecular medicine toward more personalized, precise, and effective approaches to healthcare.

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Author Contributions

H.T., Y.Z., and Y.W. contributed equally. The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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